Ex 3b: Model Planning and Building – Regression

Question

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence and Data Science
V Semester B. Tech AIDS
AD23532 – Principles of Data Science

Academic Year: 2026-2027 **Batch: 2024-2028**

Ex3b: Model Planning and Building - Regression

Learning Objective:
The objective of this experiment is to enable students to plan, build, and evaluate regression models using a real-world dataset.

Problem Statement:
A marketing agency wants to help a retail company optimize its advertising budget. The company invests in three advertising channels—TV, Radio, and Newspaper—and wants to understand how these advertising expenditures influence product sales. By developing predictive models, the company aims to estimate future sales based on advertising investments and identify the most suitable regression model for accurate sales prediction.

As a Data Scientist, your task is to build and compare multiple regression models using the Advertising dataset. Evaluate the performance of each model using appropriate regression metrics and communicate the findings through visualizations to support business decision-making.

Tasks to Perform:

1. Download the Advertising Dataset from Kaggle and load it into a Pandas DataFrame.
2. Explore the dataset by: Displaying the first and last five records, Checking the dataset shape and data types, Identifying missing values and duplicate records, Displaying summary statistics.
3. Select: TV, Radio, Newspaper as Input Features (X), Sales as Target Variable (y).

Perform Exploratory Data Analysis:

4. Generate a Pair Plot for the selected features.
5. Plot a Correlation Heatmap.
6. Identify the feature most strongly correlated with Sales.

Model Planning and Model Building

7. Split the dataset into Training (80%) and Testing (20%) dataset.
8. Build a Linear Regression model and predict Sales for the test dataset.
9. Build a Polynomial Regression model (Degree = 2) and predict Sales for the test dataset.
10. Build a Support Vector Regression (SVR) model with the RBF kernel and predict Sales for the test dataset.

Model Evaluation

11. Evaluate all three models using: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R^2 Score
12. Compare the performance of all models using a single grouped bar chart.
13. Plot Actual vs Predicted Sales for the best-performing model.

Model Interpretation

14. Identify the model with the lowest prediction error.
15. Suggest **three recommendations** to help the company optimize its advertising budget based on the model results.

Aim

To build Linear, Polynomial (degree 2) and Support Vector regression models on the Advertising dataset to predict Sales from TV, Radio and Newspaper spending, and to identify the most accurate model.

Procedure

1. Download the Advertising dataset from Kaggle and load it into a DataFrame.
2. Explore the data: first and last five rows, shape, data types, missing values, duplicates and summary statistics.
3. Select TV, Radio and Newspaper as X and Sales as y.
4. Draw a pair plot and a correlation heatmap; identify the feature most correlated with Sales.
5. Split the data 80:20.
6. Build Linear Regression, Polynomial Regression (degree 2) and SVR (RBF kernel) models and predict test Sales.
7. Compute MAE, MSE, RMSE and R^2 for each model; plot a grouped bar chart of the metrics.
8. Plot Actual vs Predicted Sales for the best model and give three budget recommendations.

Code

```
#Mohammed Fayyad(241801165)
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

df = pd.read_csv("Advertising.csv")
df = df.drop(columns=["Unnamed: 0"], errors="ignore")
print(df.head()); print(df.tail()); print(df.shape); print(df.dtypes)
print(df.isnull().sum()); print("Duplicates:", df.duplicated().sum())
print(df.describe())

X = df[["TV", "Radio", "Newspaper"]]
y = df["Sales"]

sns.pairplot(df); plt.show()
sns.heatmap(df.corr(), annot=True, cmap="coolwarm"); plt.show()
print(df.corr()["Sales"].sort_values(ascending=False))
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

models = {
    "Linear": LinearRegression(),
    "Polynomial (deg 2)": make_pipeline(PolynomialFeatures(2), LinearRegression()),
    "SVR (RBF)": make_pipeline(StandardScaler(), SVR(kernel="rbf", C=100)),
}

rows, preds = [], {}
for name, m in models.items():
    m.fit(X_train, y_train)
    p = m.predict(X_test); preds[name] = p
    mse = mean_squared_error(y_test, p)
    rows.append([name, mean_absolute_error(y_test, p), mse, np.sqrt(mse),
r2_score(y_test, p)])

res = pd.DataFrame(rows, columns=["Model", "MAE", "MSE", "RMSE", "R2"])
print(res)
res.set_index("Model").plot(kind="bar", figsize=(9, 5)); plt.xticks(rotation=0);
plt.show()

best = res.sort_values("RMSE").iloc[0]["Model"]
print("Lowest error:", best)
plt.scatter(y_test, preds[best]);
plt.plot([y.min(), y.max()], [y.min(), y.max()], "r--")
plt.xlabel("Actual Sales"); plt.ylabel("Predicted Sales"); plt.title(best);
plt.show()
```

Result

The three regression models were trained and compared using MAE, MSE, RMSE and R^2. The model with the lowest prediction error was identified, its Actual vs Predicted plot was drawn, and budget-allocation recommendations were made based on the correlation of each channel with Sales.

FIRST FIVE ROWS

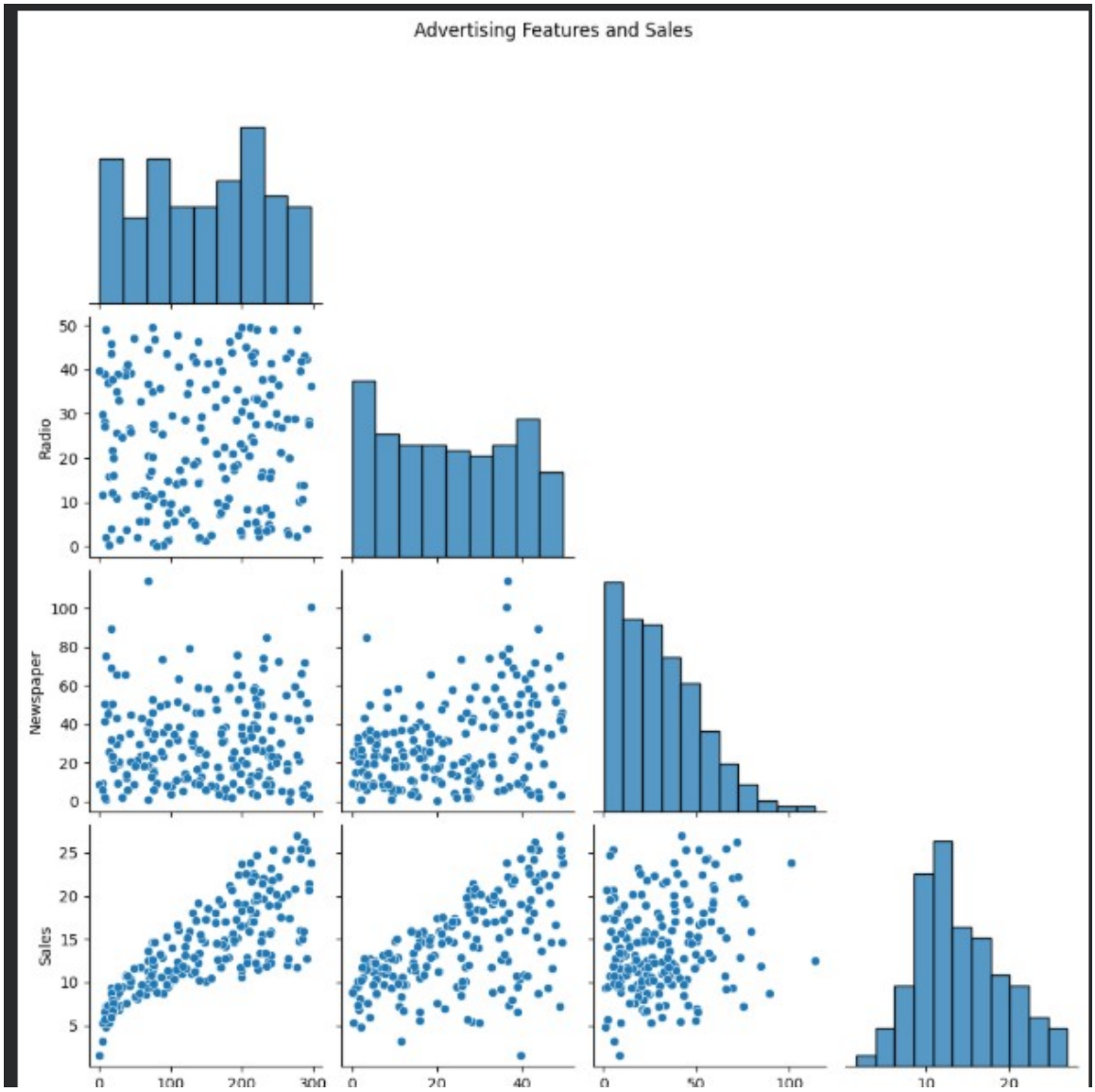
	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	9.3
3	151.5	41.3	58.5	18.5
4	180.8	10.8	58.4	12.9

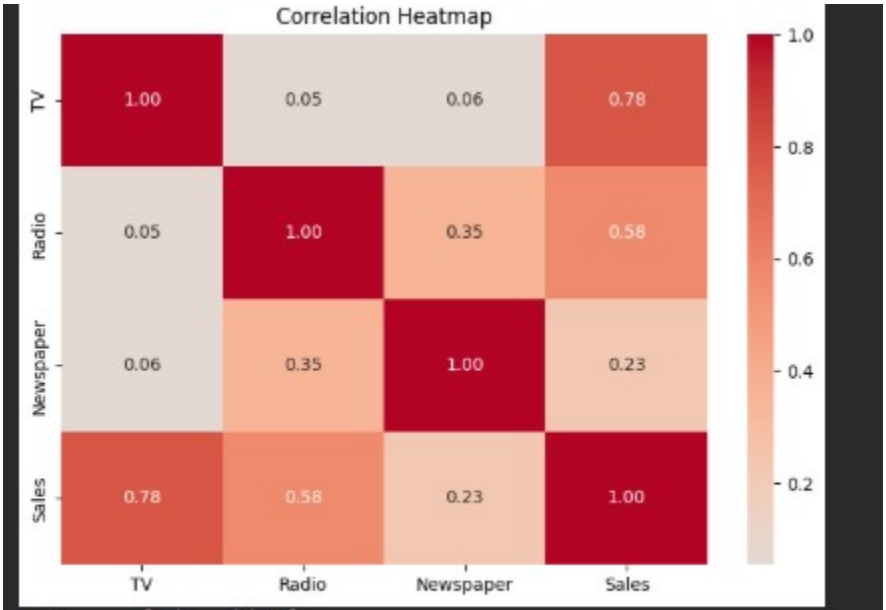
LAST FIVE ROWS

	TV	Radio	Newspaper	Sales
195	38.2	3.7	13.8	7.6
196	94.2	4.9	8.1	9.7
197	177.0	9.3	6.4	12.8
198	283.6	42.0	66.2	25.5
199	232.1	8.6	8.7	13.4

```
Shape: (200, 4)
Data types:
 TV           float64
Radio         float64
Newspaper     float64
Sales         float64
dtype: object
Missing values per column:
 TV           0
Radio         0
Newspaper     0
Sales         0
dtype: int64
Duplicate rows: 0
Summary statistics:
```

	TV	Radio	Newspaper	Sales
count	200.00	200.00	200.00	200.00
mean	147.04	23.26	30.55	14.02
std	85.85	14.85	21.78	5.22
min	0.70	0.00	0.30	1.60
25%	74.38	9.98	12.75	10.38
50%	149.75	22.90	25.75	12.90
75%	218.82	38.52	45.10	17.40
max	296.40	49.60	114.00	27.00

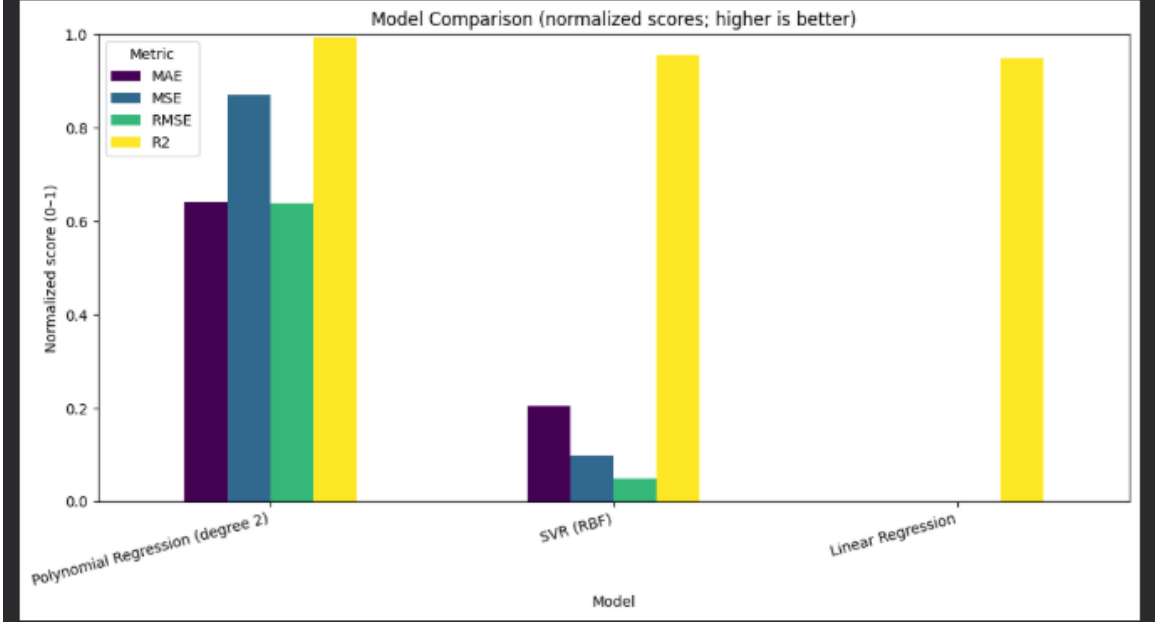

Advertising Features and Sales
Radio
Newspaper
Sales

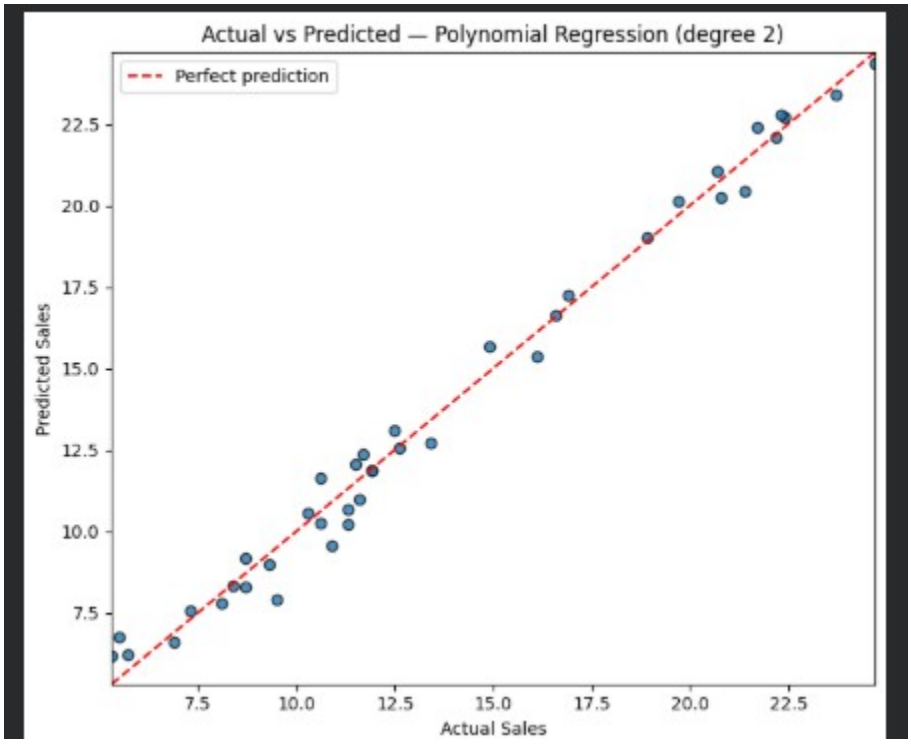


TEST-SET MODEL METRICS (lower errors and higher R2 are better)

Model	MAE	MSE	RMSE	R2
Polynomial Regression (degree 2)	0.5262	0.4129	0.6426	0.9889
SVR (RBF)	1.1621	2.8621	1.6918	0.9093
Linear Regression	1.4608	3.1741	1.7816	0.8994

Lowest prediction error (by RMSE): Polynomial Regression (degree 2)




Actual vs Predicted — Polynomial Regression (degree 2)
Perfect prediction
Predicted Sales
Actual Sales
7.5
10.0
12.5
15.0
17.5
20.0
22.5